Shibo Yan

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https://yan-shibo.github.io



Education

Southwest University | Software Engineering | Full-time
College of Computer and Information Science & College of Software (the College)

Master's Program (expected graduation) Sep 2024 - Jun 2027

Bachelor's Degree Sep 2019 - Jun 2023

GPA: 3.95/5.00; rank in major: 8/124. Course grades: Data Structures 98, Probability Theory 98, Data Mining 94, Software Modeling and Design 96.

Research Experience

Trustworthy Software Laboratory, Southwest University | Master's Student Sep 2024 - Jun 2027

Research: learning-aided reach-avoid controller synthesis with formal probabilistic guarantees for nonlinear stochastic dynamical systems.

- Participate in a National Natural Science Foundation of China General Program project, "Polynomial Algebraic Methods for Safety Verification of Neural-Network-Controlled Systems" (No. 62472362; Jan 2025 - Dec 2028).
- *Learning-Aided Reach-Avoid Controller Synthesis with Formal Guarantees for Stochastic Systems*. MEMOCODE 2026 (CCF C conference). **First author; accepted.**
- *Safety Guarantees for Nonlinear Systems with Bayesian Neural Network Control Policies*. *Formal Aspects of Computing* (CCF B journal). **Second author (supervisor first); submitted.**
- *Formal Reach-Avoid Controller Synthesis for Stochastic Systems via Iterative Neural-Symbolic Learning*. CCF A conference. **Second author (supervisor first); submitted.**

Projects and Internship Experience

CBF-Rover | Mobile Robot Safe Control Simulation | Independent Developer Mar 2025 - Present

Develop system models, high-level architecture, and barrier-certificate methods for stochastic disturbances and dynamic obstacles. Implemented certificate synthesis, independent verification, and online safety filtering using control barrier functions (CBFs).

Facial Recognition Attendance System | Undergraduate Capstone Dec 2022 - Jun 2023

Second Prize, University Outstanding Undergraduate Thesis Award. Designed and built a multi-platform system using face detection and recognition to identify attending students from captured or uploaded images and generate attendance summaries.

Chongqing Chinasoft International Information Technology Co., Ltd. Jun 2022 - Aug 2022

Internship Project Team Technical Lead & Project Manager | Outstanding Intern. Coordinated team development and integration of **MicFamily**, a karaoke venue operations management system, using Spring Boot, MySQL, and uni-app; internship score: 94; code released on GitHub.

More projects: GitHub | Personal website

Honors and Awards

Second Prize, "Huawei Cup" China Post-Graduate Mathematical Contest in Modeling 2024

Honourable Mention, 9th "Certification Cup" International Mathematical Modeling Competition 2020

Second Prize, 13th and 14th National College Student Mathematics Competition (Chongqing Division) 2021 - 2022

Undergraduate: Second-Class Scholarship; Merit Student; Civic Conduct Award; Outstanding Communist Youth League Member 2019 - 2022

Graduate: Outstanding Individual in Academic and Technological Innovation; First-Class Academic Scholarship 2024 - 2025

Additional competitions and awards: Personal website.

Leadership and Activities

TIPCC, Tsinghua University Press | Appointed Volunteer May 2022 - May 2024

National Higher-Education Expert Committee on Computing and Industry Practice Resource Development

Shapingba District Youth Talent Grassroots Service Program Sep 2021 - Aug 2022

Publicity and Planning Team Lead | University Outstanding Project, 2021 - 2022

The College: Student Union | Academic Affairs & Technology Department Member Sep 2019 - May 2020

The College, Southwest University | Research Assistant Jun 2023 - Jun 2024

Baidu Songguo Elite Program | Outstanding Program Graduate; ranked 2nd by total points Sep 2021 - Dec 2022